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Resource Management in a Multi-User Linux-Based Server Environment

CO1 AT3 Technical Presentation
CSA0401 Operating Systems

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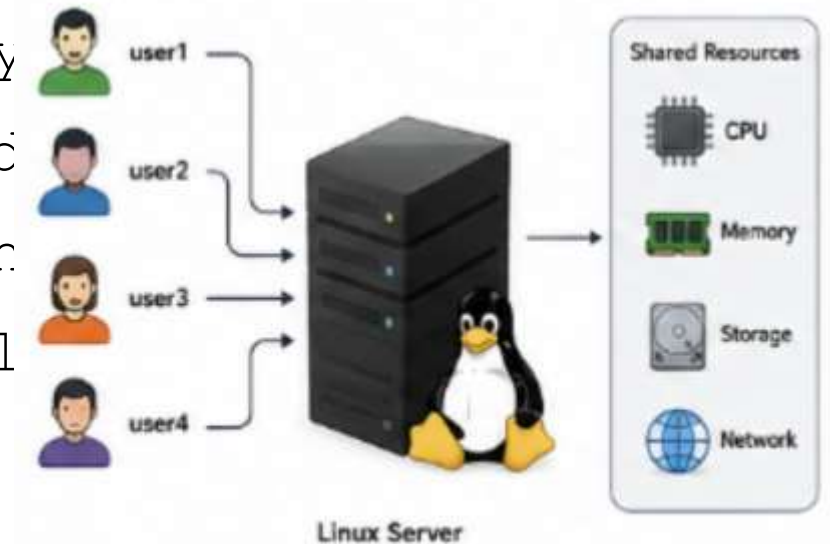
Introduction

What is a Multi-User Linux Server?

- Supports multiple users simultaneously
- Shares files, printers, storage, and c
- Uses process scheduling and memory mar
- Ensures security and fair resource all

Examples

- College servers
- Cloud servers
- Web hosting servers



OS Functions Involved in Resource Management

The Operating System manages all hardware and software resources.

Process Management

- Creates processes
- Schedules CPU
- Terminates processes

Memory Management

- Allocates RAM
- Virtual memory
- Swapping

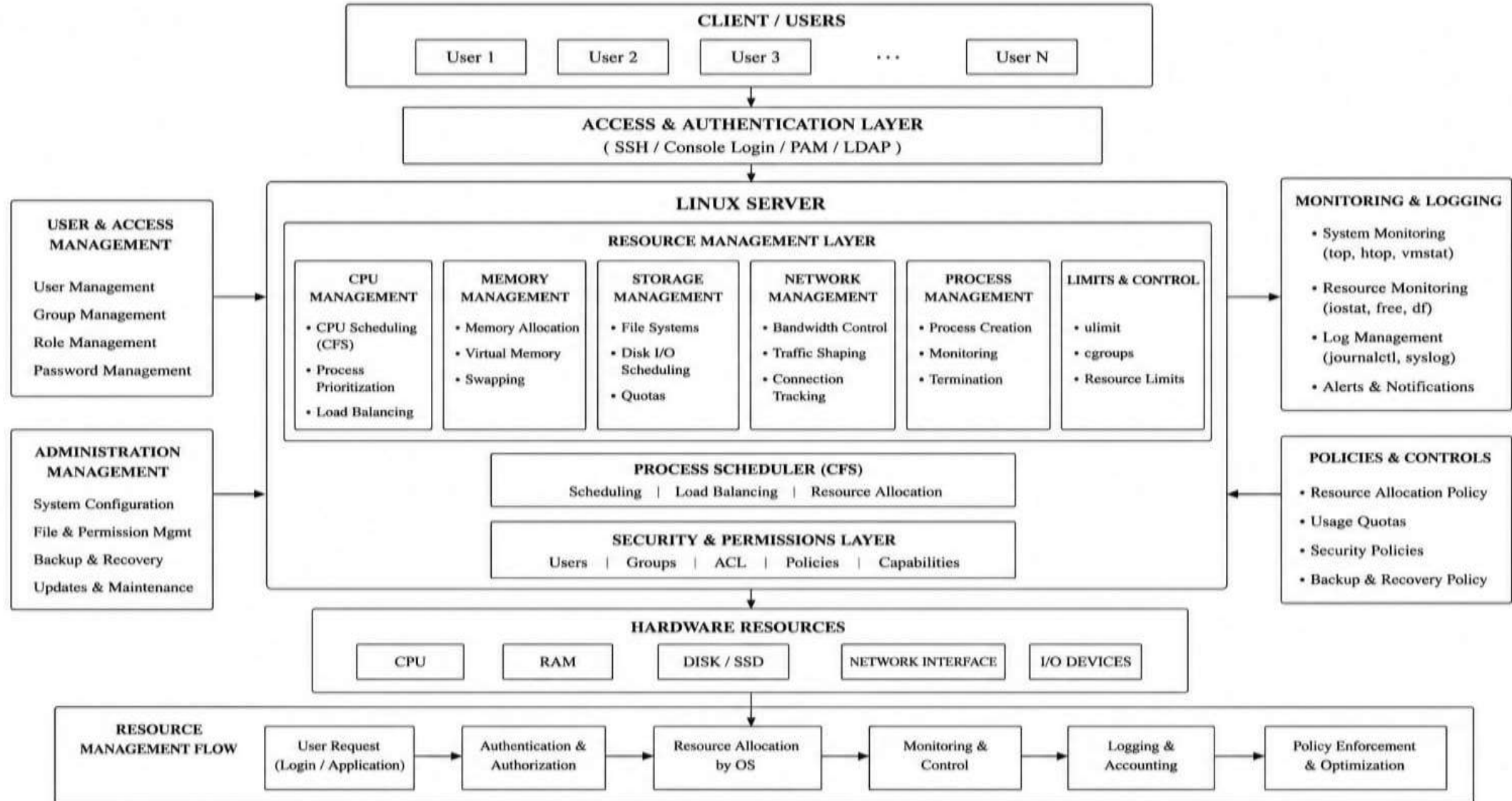
File Management

- Creates files
- Deletes files
- Controls permissions

Device Management

- Controls printers
- Hard disks
- USB devices
- Keyboard and

Resource Management Architecture



File System Calls

System calls allow user programs to access files.

Example

```
fd = open("sample.txt", O_RDONLY);  
read(fd, buffer, 100);
```

System Call	Purpose
open()	Opens a file
read()	Reads data
write()	Writes data
close()	Closes file
lseek()	Moves file pointer
stat()	Gets file information
mkdir()	Creates directory
unlink()	Deletes file

Device System Calls

Linux treats devices as files.

Common Device Calls

Examples

- Keyboard
- Mouse
- USB Drive
- Printer
- Hard Disk

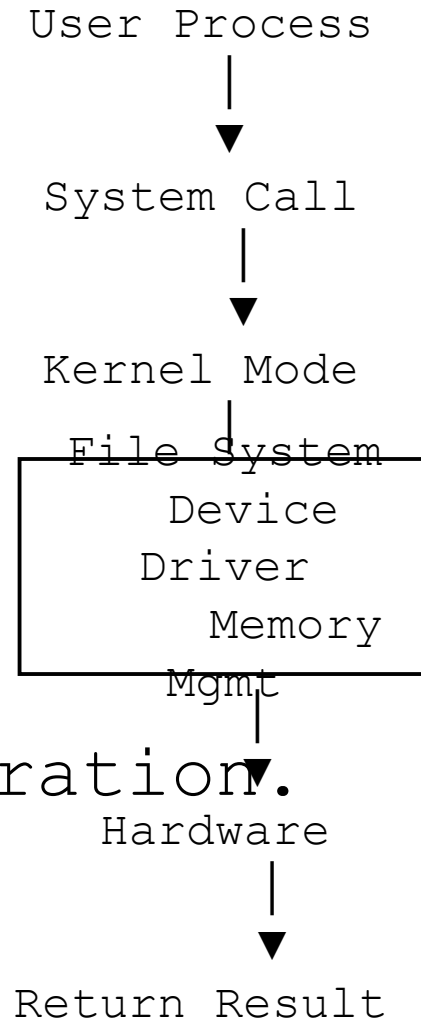
System Call	Function
<code>open()</code>	Open device
<code>read()</code>	Read device
<code>write()</code>	Send data
<code>ioctl()</code>	Device control
<code>close()</code>	Close device

Interaction Between User Processes and Kernel

Step-by-Step

- User program requests a service.
- It invokes a system call.
- CPU switches to kernel mode.
- Kernel validates permissions.
- Kernel performs the requested operation.
- Control returns to user mode.

Flow Diagram



Advantages

- Efficient CPU scheduling
- Secure file sharing
- Controlled device access
- Reliable multitasking
- Better memory utilization



Conclusion

- Linux efficiently manages shared resources.
- System calls provide communication between applications and the kernel.
- Resource management ensures security, fairness, and high performance in multi-user environments.

THANK YOU